M&AE 4070/5070 - Dynamics of Flight Vehicles

Chapter 2 - Longitudinal Equilibrium

Companion Notes for Case Study 2

This chapter introduces the longitudinal equations of motion, that is, motion of the aircraft that is fully contained in a vertical plane. It then restricts the focus to (pseudo)-equilibrium cases, for which time dependency can be neglected. But before looking at the equations themselves, let's go over naming conventions that will be used throughout the course, define clearly our options in terms of reference frames, and go over some aircraft nomenclature.

1 Naming conventions

The following table details the most likely notations used throughout the course.

	Vector components				
	Acceleration	Velocity	Position	Moments/Forces	Controls
Linear	a_x, a_y, a_z [m/s²]	u, v, w	x, y, z	X, Y, Z	δ_x
Rotation	a_p, a_q, a_r [rad/s²]	angular veloc. p, q, r [rad/s]	ψ, θ, φ	L, M, N	$\delta_\ell, \delta_m, \delta_n$

forces; throttle; roll; angles; heading; pitch; roll/bank; moments; pitch; yaw

Note: Thrust forces are denoted F_X, F_Y, F_Z, while force and moment coefficients all start with C.

(yes, F, not T)

2 Overview of reference frames

From experience, the system of equations governing the motion of an object is often simpler in a referential close to the object under consideration. For an aircraft, what is considered "close" depends on the context and which specific motion is of interest. Overall, three different frames of reference are relevant and will be used throughout the semester. They are simply listed here, but will be detailed later in the relevant chapters.

"ABSOLUTE" frame of reference + 2 "AIRCRAFT"-related frames of reference

Earth reference
Subscript "0"

Aircraft reference
Subscript "b"

Aerodynamics reference
Subscript "a"

Earth-based frame of reference $R_o\ [\vec{x_o}, \vec{y_o}, \vec{z_o}]$

- Origin O is a fixed (arbitrary) point on the ground
- $\vec{z_o}$: in the direction of the gravitational force seen by the plane ⟹ pointing DOWN
- $\vec{x_o}$: horizontal, pointing toward true or magnetic North
- $\vec{y_o}$ completes the basis vectors according to the right hand rule.

Aerodynamic frame of reference $R_a\ [\vec{x_a}, \vec{y_a}, \vec{z_a}]$

- Origin G, same as aircraft-based reference frame ⟹ G ≡ center of gravity of aircraft
- $\vec{x_a}$: oriented in the direction of the aircraft velocity (pointing away from aircraft)
- $\vec{y_a}$ and $\vec{z_a}$: obtained by rotation of the aircraft-based reference frame by 2 angles, α (angle of attack) and β (side-slip angle, assume to be zero in this chapter)

Aircraft-based frame of reference R_b $[\vec{x}_b, \vec{y}_b, \vec{z}_b]$

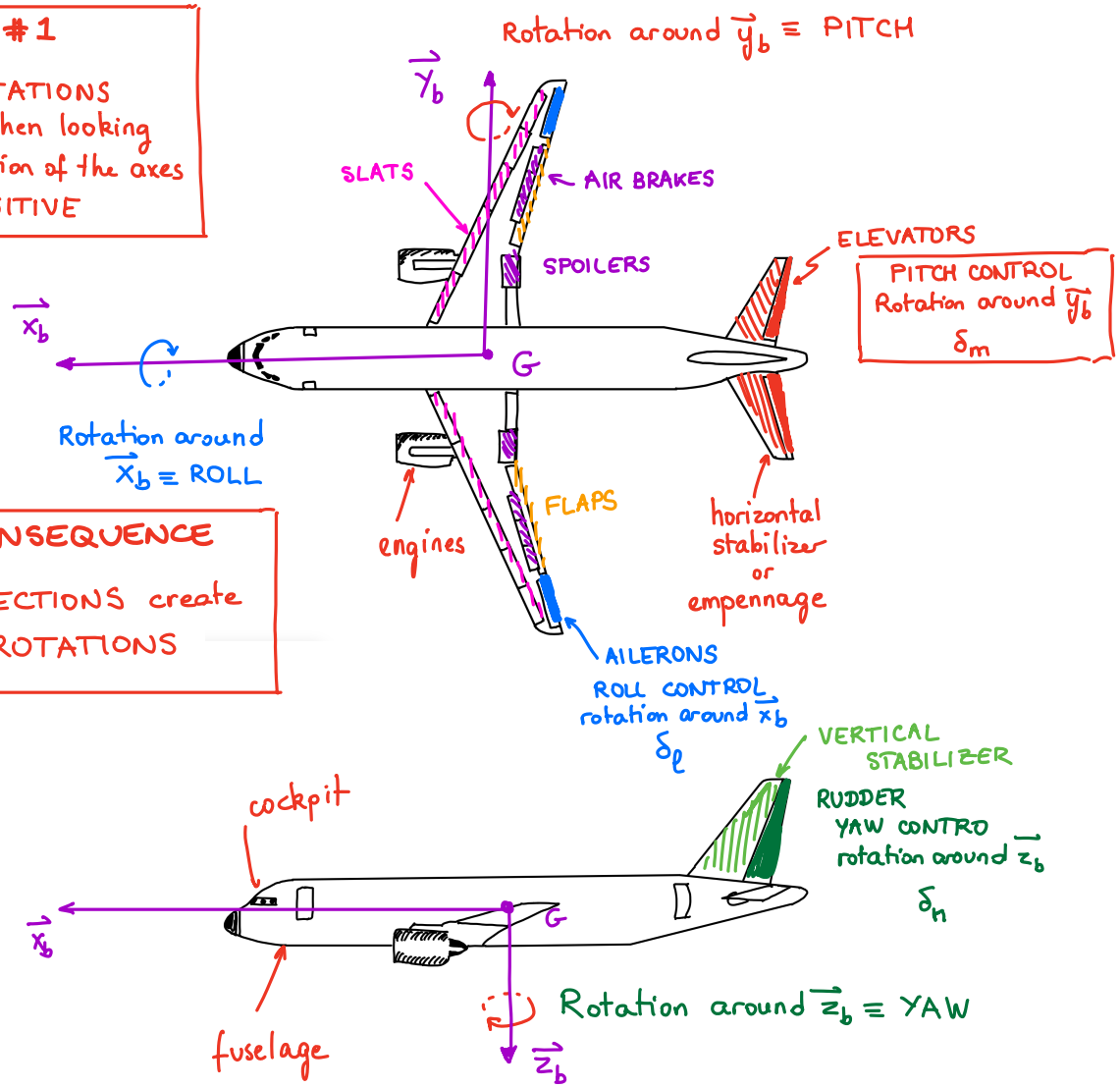
- Origin G, center of gravity of the plane (even if not technically fixed due to fuel consumption)
- $\vec{x_b}$: oriented forward, along the fuselage axis
- $\vec{z_b}$: in the plane of symmetry of the aircraft, oriented toward the "floor" of the aircraft
- $\vec{y_b}$: perpendicular to the plane of symmetry of the aircraft, oriented towards the pilot's right.

3 Controls, control surfaces, and conventions

REF FRAME
Aircraft frame of reference R_b

CONVENTION #1

CLOCKWISE ROTATIONS AROUND AXES when looking in the positive direction of the axes are counted as POSITIVE



DIRECT CONSEQUENCE

POSITIVE DEFLECTIONS create NEGATIVE ROTATIONS

PITCH δ_m
ROLL δ_ℓ
YAW δ_n
} angles, can go ≥ 0 and ≤ 0

[note: the sign is decided by deflection from aileron on the RIGHT side of the pilot.]

THROTTLE $0 \leq \delta_x \leq 1$ (0 to 100%)

Note: Except for throttle, which is associated with the thrust force, the controls exert moments on the aircraft

4 Decoupling longitudinal and lateral equations of motion

The full system of equations describing the motion of an aircraft contains 6 second-order ordinary differential equations (3 for translation, 3 for rotation) of the form:

$$m\,\ddot{\text{position}} = \Sigma\,\text{forces} \quad (\text{aero} + \text{propulsion})$$

$$I\,\ddot{\text{angular position}} = \Sigma\,\text{moments}$$

$\ddot{}$ = 2nd time derivative; I = matrix/moment of inertia

We convert them into 12 **first-order** ODEs: 6 so-called "mechanical" equations of the form:

$$\begin{cases} m\,\dot{\text{velocity}} = \Sigma\,\text{forces} \\ I\,\dot{\text{angular velocity}} = \Sigma\,\text{moments} \end{cases}$$

and 6 "kinematic" equations of the form:

$$\begin{cases} \dot{\text{position}} = \text{velocity} \\ \dot{\text{angular position}} = \text{angular velocity} \end{cases}$$

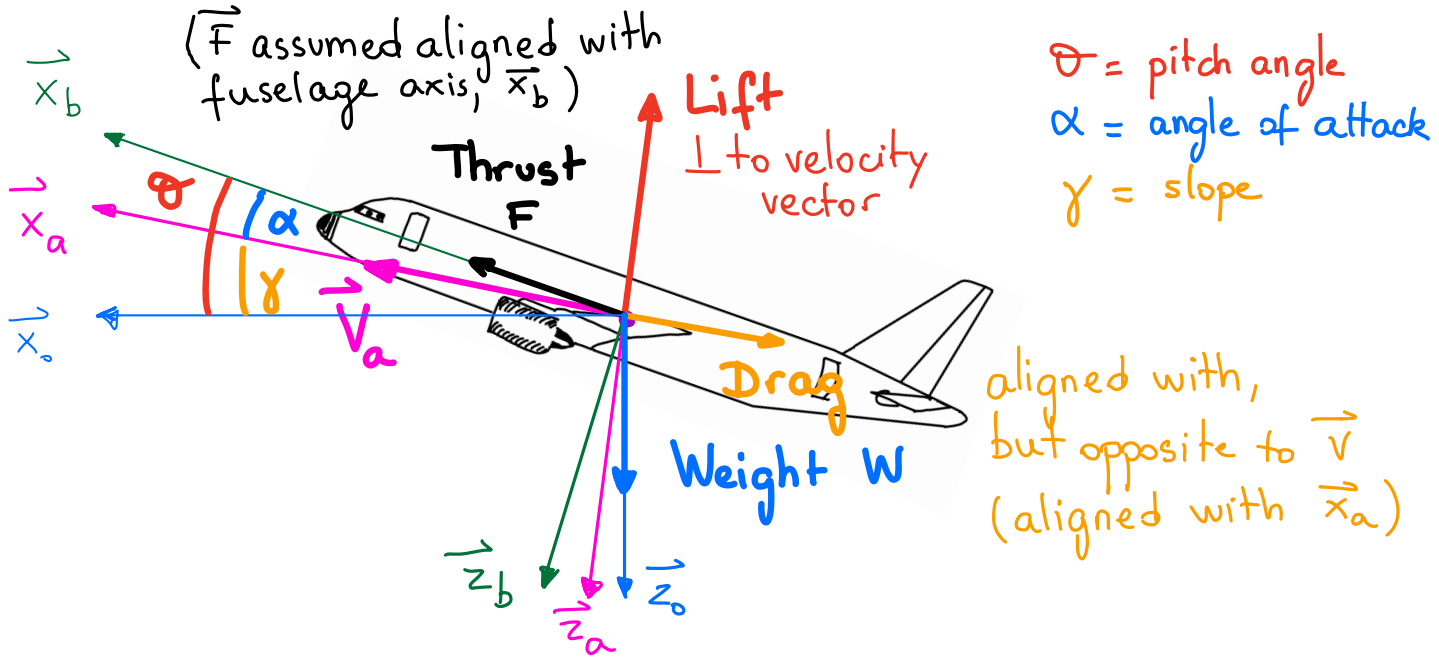
Because the atmosphere is isotropic horizontally (invariant under an horizontal translation), the geographical position of the aircraft (x_G, y_G) and its bearing ψ have no influence on forces and moments, reducing the number of meaningful equations from 12 to $12 - 3 = 9$. Decoupling the longitudinal motion from the lateral one splits the equations into two groups: 5 for the longitudinal motion, 4 for the lateral motion. For this, a number of assumptions need to be made to constrain the motion of the aircraft onto the vertical plane:

1. Weight must be vertical.
2. All forces (aerodynamics and propulsion) must be in or parallel to the plane of symmetry of the aircraft.
3. All moments (from aerodynamic and propulsion forces) must be perpendicular to the plane of symmetry of the aircraft.

That implies:

- There is a vertical plane of symmetry, wings are horizontal
  - ↳ roll angle $\varphi = 0$
- Velocity vector has to belong to the vertical plane
  - ↳ side-slip angle (see later) = $\beta = 0$
- Roll and yaw angular velocities are 0 : $p = r = 0$

R_a is the most convenient reference frame to look at longitudinal motion, and therefore, the longitudinal equations of motion should be expressed in R_a:



Earth frame of reference R_o
Aero frame of reference R_a
Aircraft frame of reference R_b

Note: only the weight is strictly applied at G, Lift, drag and thrust are not.

BUT we can assume that drag and thrust are ≈ applied at G
Lift is applied at what is called the aerodynamic center of the aircraft.

DISCLAIMER. The complete, general equations of motion for an object moving in three-dimensional space are tedious to derive. Instead, we will rely on a number of implicit assumptions to quickly obtain simplified equations appropriate for conventional flight. The flip side of this is that we won't be able to look at the aerodynamics of fighter jets doing fancy maneuvers.

- Note that the equations we will be using CAN be derived rigorously from the complete, rigorous ones.
- Example of implicit assumptions: neglecting extra terms coming from the rotation of R_a relative to R_o.
- Full set of equations can be found in appropriate texts, *e.g. Flight Dynamics* by Stengel.

The full system of longitudinal equations is written:

Variable

- Force balance along $\vec{x}_a$ [P] Propulsion equation

$$m\dot{V} = -\text{Drag} + F\cos\alpha - mg\sin\gamma \quad (1) \quad [V]$$

($\cos\alpha \approx 1$, $\sin\gamma \approx \gamma$)

Need models to quantify those terms (Chapter 1, MAE 3050)

$$m\dot{V} = -\frac{1}{2}\rho S V^2 C_x + F\cos\alpha - mg\sin\gamma$$

- Force balance along $\vec{z}_a$ [S] Sustention equation

$$mV\dot{\gamma} = -\text{Lift} + F\sin\alpha + mg\cos\gamma \quad (2) \quad [\gamma]$$

($\sin\alpha \approx \alpha$, small; $\cos\gamma \approx 1$; Lift $\gg$ thrust)

See end of chapter to see quick and dirty way to understand where this term is coming from

$$mV\dot{\gamma} = -\frac{1}{2}\rho S V^2 C_z + F\sin\alpha + mg\cos\gamma$$

- Moment balance around $\vec{y}_a$ [M] Moment equation (around center of gravity)

$$B\dot{q} = M_{aero} + M_F \quad (3) \quad [q]$$

(M_F: moment created by thrust → neglect it in this class)

(B: moment of inertia around $\vec{y}_a$)

$$B\dot{q} = \frac{1}{2}\rho S V^2 \ell C_m + M_F$$

- Kinematic equation for the pitch velocity around $\vec{y}_a$ [Kq]

$$\dot{\alpha} + \dot{\gamma} = q \quad (4) \quad [\alpha]$$

- Altitude kinematic equation along $\vec{z}_o$ [Kh] altitude equation

$$\dot{h} = V\sin\gamma \quad (5) \quad [h]$$

($\sin\gamma \approx \gamma$)

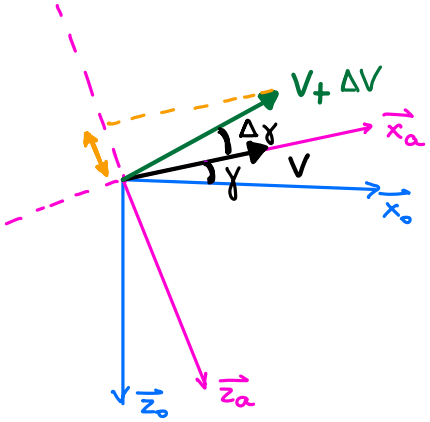
In those equations:

- α is the angle of attack of the aircraft, defined as the the angle between the fuselage axis ($\overrightarrow{x_b}$) and the projection of the velocity vector of the aircraft on the symmetry plane of that aircraft.
- B is a term that describes in which ratio main wings and empennage contribute to the moment. It will be detailed in the next chapter.
- ℓ is a reference length, taken as the mean chord for longitudinal equations.
- F is the magnitude of the thrust force.
- M_{F_G} is the moment around the center of gravity created by the thrust force.

Most of the time, the following simplifications are made:

1. $\overrightarrow{F}$ is parallel to $\overrightarrow{V}$. This assumes that the component of $\overrightarrow{F}$ perpendicular to $\overrightarrow{V}$ is negligible compared to the other forces involved, which is valid especially at high velocity and low angle of attack. **Decouples $\boxed{\text{P}}$ and $\boxed{\text{S}}$ with respect to F.**
2. $M_{F_G} = 0$. This assumes that the thrust force goes through the center of gravity. True for a fighter jet, less true for commercial airplane with high or low engines. **Decouples $\boxed{\text{P}}$ and $\boxed{\text{M}}$ with respect to F.**
3. Moderate slope γ ($\cos\gamma \approx 1$, and $\sin\gamma \approx \gamma$). True, except for fighter jets doing crazy things. **Decouples $\boxed{\text{P}}$ and $\boxed{\text{S}}$ with respect to γ.**

Quick note on where the acceleration term along $\overrightarrow{z_a}$ is coming from



Consider change in velocity over small time interval Δt

V becomes $V + \Delta V$

γ becomes $\gamma + \Delta\gamma$

$\Rightarrow$ neglecting the fact that $\vec{R}_a$ also rotates by $\Delta\gamma$

$$\text{Acceleration along } \vec{z}_a = \frac{\text{change in velocity along } \vec{z}_a}{\Delta t}$$

$$= \frac{\overbrace{(V+\Delta V)\sin\Delta\gamma}^{\text{Vel. projected on } z_a \text{ after } \Delta t} - \overbrace{0}^{\text{Velocity is initially } \perp \text{ to } \vec{z}_a}}{\Delta t}$$

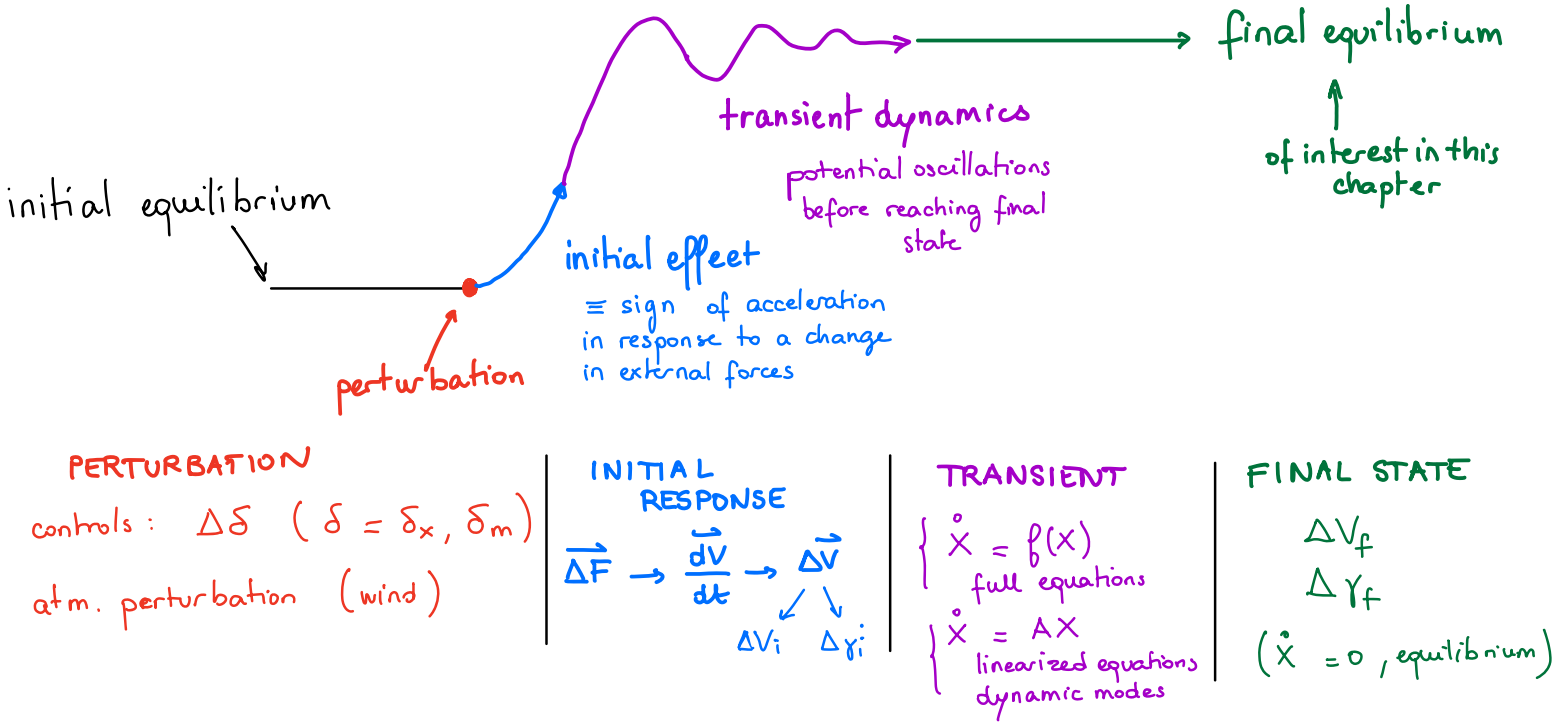
$$\approx \frac{V\Delta\gamma}{\Delta t} \xrightarrow{\Delta t \to 0} V\dot{\gamma}$$

$\sin\Delta\gamma \sim \Delta\gamma$ for small $\Delta\gamma$

$\Delta V\,\Delta\gamma \ll V\Delta\gamma$

5 Response of a mechanical system to a perturbation

In this chapter, we are interested in equilibrium flight conditions, which we also will call interchangeably steady-state conditions, and how a given equilibrium state is impacted by controls. First, we clarify where the steady-state fits in the general framework of a mechanical system (aircraft) being disturbed by a perturbation (control or external forces). This response can be decomposed into 3 successive (in time) stages:



A few comments:

- A final equilibrium is obtained **only if the system is stable**
- For a linear, stable system, the final equilibrium is different from the initial equilibrium **only if the perturbation is permanent**: if the perturbation goes away, the system relaxes back to the initial equilibrium.
- The aircraft being reasonably well modeled by a system of linear, stable equations, **one set of controls will lead to one, and one only, equilibrium state.**

One rudimentary control strategy therefore could consist in setting the controls directly to the position corresponding to the desired flight equilibrium state, and just wait until that state is reached by the aircraft alone. This strategy, however, is definitely not a great one if the dynamics part of the aircraft response involves a very slow mode that loops back to the controls (more about that later).

Initial effects of $\Delta\delta_x$ and $\Delta\delta_m$

The initial and transient behavior of an aircraft to a change in pitch control and throttle will be covered in more detail later on. Only a brief overview is given here, as it is relevant for the study of the steady-state equations.

- **Change in throttle, $\Delta\delta_x$:** $m\dot{V} = -\frac{1}{2}\rho S V^2 C_x + F - mg\gamma$

Recall the complete equation [P]. If the thrust F is increased, the acceleration term $\dot{V}$ is positive, leading to an initial increase in velocity.

$\Delta\delta_x > 0$ (increase in throttle) $\Rightarrow$ $\Delta F > 0$ (increase in thrust) $\Rightarrow$ $\dot{V} > 0$ $\Rightarrow$ $\Delta V > 0$

δ_x impacts velocity initially

- **Change in pitch control, $\Delta\delta_m$:**

Recall the complete equation [M]. A negative $\Delta\delta_m$ will create a positive pitching moment (pitch up) and therefore, a pitching acceleration. If we were to observe the behavior of an aircraft in this situation, we could see that it goes through a rapid, well-damped oscillation of its angle of attack, quickly reaching a new, larger, angle of attack.

In turn, this nearly immediate change of angle of attack can be considered as a perturbation itself, impacting the sustentation equation [S]: lift is increased, creating a positive acceleration in $\dot{\gamma}$, and an increase in slope.

↳ $mV\dot{\gamma} = -\frac{1}{2}\rho S V^2 C_z + mg$

$\Delta\delta_m < 0$ (aircraft nose goes up) $\longrightarrow$ Dynamic motion
↳ oscillation in angle of attack
↳ damped very quickly
$\longrightarrow$ $\Delta\alpha > 0$

$\Delta\alpha > 0 \longrightarrow \Delta C_z > 0 \longrightarrow \Delta\text{Lift} > 0 \longrightarrow \dot{\gamma} > 0 \longrightarrow \Delta\gamma > 0$

$\Delta\delta_m$ impacts slope initially

In the rest of this chapter, we ignore the potential complications generated by the transient response, and focus directly on the final equilibrium state created by a permanent change in a given control (δ_m and δ_x).

6 Equations for longitudinal equilibrium and extension to pseudo-equilibrium

In this section, the concepts of equilibrium and pseudo-equilibrium (also called steady-state and pseudo steady-state) are introduced, along with the corresponding equations, and we start looking at the impact of controls, in this case pitch and throttle, on those equilibria/pseudo-equilibria, especially in terms of trajectory, that is, velocity and slope.

We make the assumption the slope γ remains small. At equilibrium/steady state, all temporal derivatives are set to zero:

$$\dot{V} = \dot{\gamma} = \dot{q} = \dot{\alpha} = \dot{h} = 0 \tag{6}$$

- **P** along $\vec{x_a}$ (α small $\Rightarrow \cos\alpha \approx 1$; γ small $\Rightarrow \sin\gamma \approx \gamma$) $\quad F = \frac{1}{2}\rho S V^2 C_x + mg\gamma$
- **S** along $\vec{z_a}$ (γ small $\Rightarrow \cos\gamma \approx 1$) $\quad mg = \frac{1}{2}\rho S V^2 C_z$
- **M** around/along $\vec{y_a}$ $\quad C_m = 0$ $\quad$ no pitching moment because $\dot{q} = 0$
- **Kq** $\quad q = 0$
- **Kh** along $\vec{z_0}$ $\Rightarrow \quad \dot{h} = V\sin\gamma \approx V\gamma = 0$

A few comments:

1. **Kq** imposes a zero pitch velocity.The aircraft AND the velocity vector do not rotate in the vertical plane: the trajectory is a straight line.

   Longitudinal flight at steady state is along a straight line
   $\Rightarrow q = 0$, α = constant

2. **Kh** imposes $V \sin\gamma = 0$: the velocity is zero (not desirable for the aircraft) or the slope is zero

   Longitudinal flight at steady state is along straight, horizontal line : $\gamma = 0$
   ↳ technically, ascent/descent are NOT steady state flight configurations
   ↳ to account for them, we introduce pseudo-equilibrium configurations where $\dot{h} \neq 0$ but $\dot{V}$, $\dot{\alpha}$, $\dot{\gamma}$ and $\dot{q}$ ARE 0

   Indeed, during ascent or descent, the change in density calls into question the equilibrium or steady state of equation **S**. In practice, this unsteady contribution is minor, and just ignoring the altitude kinematic equation yields results that are fully representative of the aircraft's behavior.

   $\Rightarrow$ even if ρ is changing when h changes, the changes are small enough to assume that ρ remains constant in the **P**, **S**, **M**, and **Kq** equations.

7 Role of controls on equilibrium/pseudo-equilibrium flight

Let's look at what controls do in the context of equilibrium or pseudo-equilibrium flight. This is most conveniently done by analyzing each equation one after the other.

7.1 Moment equation [M]

The analysis of the equation of moment [M] provides the following fundamental result (the demonstration is quite lengthy, and will be kept for the next chapter):

[M] $C_m = 0$ at equilibrium or pseudo-equilibrium

* to modify the angle of attack on an aircraft, NEED TO CREATE PITCHING MOMENT
  - ↳ pitching moment can be changed by δ_m
  - ↳ position of pitch control δ_m proportional to α $\quad \boxed{\delta_m \Longleftrightarrow \alpha}$
* throttle δ_x does NOT create a pitching moment : IT CANNOT CHANGE α

$$\boxed{\delta_x \not\Longleftrightarrow \alpha}$$

7.2 Sustentation equation [S]

$$mg = \frac{1}{2}\rho S V^2 C_z$$

(C_z: a function of α only)

- When in a longitudinal steady state, lift is constant, because it is equal to weight.
- When in a longitudinal pseudo-steady state (ascent/descent), it is still true!

In equilibrium/pseudo equilibrium, LIFT IS NOT RESPONSIBLE for ascent/descent !

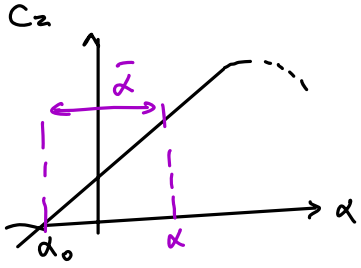
Equation without small angle approximation :

$$mg \underbrace{\cos\gamma}_{<1} = \frac{1}{2}\rho S V^2 C_z$$

⇒ As soon as the aircraft departs from level flight, lift becomes SMALLER! (it would go to 0 for vertical ascent with $\gamma = 90°$!)

⇒ ascent/descent come from projection of thrust Force on vertical direction.

Role of sustentation equation:
Assuming that density is constant (pseudo-equilibrium), [S] shows that: $V^2 C_z = \text{constant}$



$C_z \propto \alpha - \alpha_0 = \bar{\alpha}$

↳ $V^2 \bar{\alpha} = \text{constant}$

Let's take the differential form of that equation, assuming small variations: $2\frac{\Delta V}{V} = -\frac{\Delta \alpha}{\alpha}$

* δ_m controls α, and a change in α creates a change in velocity

  $\Rightarrow$ δ_m controls velocity $\quad \boxed{\delta_m \Longleftrightarrow V}$

* δ_x not being able to change α, it CANNOT change velocity !

  $\Rightarrow$ δ_x does NOT control velocil $\quad \boxed{\delta_x \not\Longleftrightarrow V}$

Important : remember that those results apply to steady or pseudo steady state, after initial effects and dynamics have died down.

those results are different from the more intuitive, "initial effect" discussion

Here, we are not concerned by HOW the aircraft reaches the new steady state.

What if we keep the stick in a fixed position, *i.e.*, keep δ_m constant?

hypothetical scenario: $\delta_x \uparrow$, keep δ_m constant : what happens to D (drag force)? (after a new equilibrium is obtained)

* [M] if δ_m is fixed, α is fixed : constant angle of attack

  ↳ if α = constant, C_z = constant because $C_z = f(\alpha)$

  ↳ if C_z = constant, C_x = constant (drag polar)

* [S] if C_z = constant, $\frac{1}{2}\rho V^2 S$ = constant (mg = constant)

so $D = \underbrace{\frac{1}{2}\rho V^2 S}_{\text{cst}} \underbrace{C_x}_{\text{const}} \Rightarrow D = \text{constant}$

$\boxed{\delta_m = \text{constant} \Rightarrow \text{DRAG} = \text{constant}}$

7.3 Propulsion equation P

thrust

$$F = \frac{1}{2}\rho S V^2 C_x + mg\gamma$$

$\frac{1}{2}\rho S V^2 C_x$: Drag

$mg\gamma$: projection of weight onto $\vec{x_a}$

* thrust F used to compensate drag + projection of weight
  - in level flight, only balances drag
  - there cannot be an ascent unless thrust larger than drag

* to find out if aircraft can climb, need to know if we can get a positive $F - D$
  - easiest way is to increase δ_x
  - if δ_m is kept constant, effect is direct (drag is constant)

$$\Delta F(\delta_x) = mg\Delta\gamma$$

THROTTLE CONTROLS γ

7.4 Propulsion P and Sustentation S equations

So far, we have seen that:

- δ_m (stick, pitch control) changes velocity.
- δ_x (throttle) changes on slope γ.
- δ_x (throttle) cannot change velocity.

The final question is to figure out if δ_m can have an effect on the slope γ. P shows that to change the slope γ, we need to change the thrust balance:

$$\Delta F = \text{thrust} - \text{drag} = T - D \tag{7}$$

Since δ_m changes the velocity, how does it modify the thrust balance, especially through the drag term, as a function of velocity?

The drag term is expressed as:

$\frac{1}{\pi AR}$ S $C_z = \frac{mg}{\frac{1}{2}\rho S V^2}$

$$D = \frac{1}{2}\rho S V^2 C_x(\alpha) = \frac{1}{2}\rho S V^2 \left[\underbrace{C_{x_0} + k C_z^2}_{\text{drag polar}}\right] \tag{8}$$

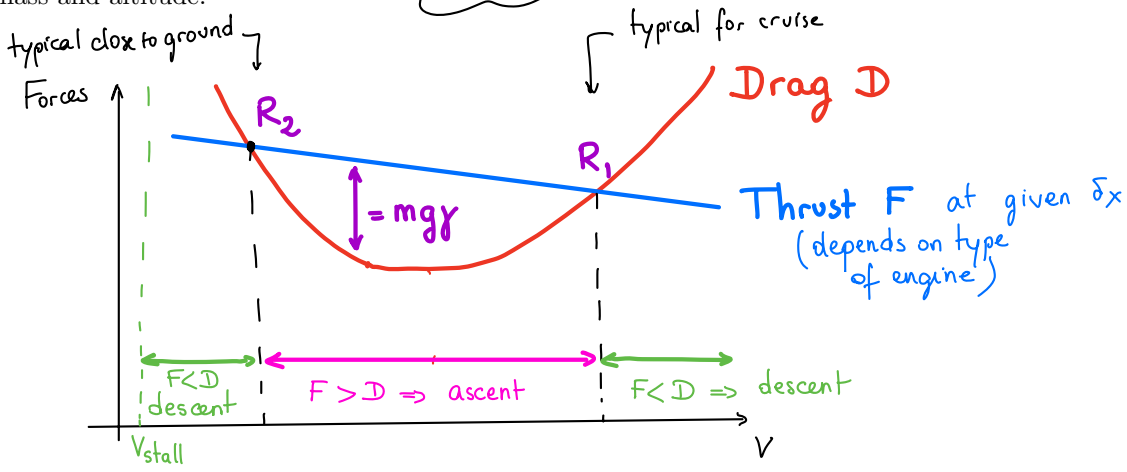
$$= \frac{1}{2}\rho S V^2 \left[C_{x_0} + \frac{k(mg)^2}{\left(\frac{1}{2}\rho S V^2\right)^2}\right]$$

of the form: $D = A V^2 + \frac{B}{V^2}$ $\rightarrow B = k \frac{(mg)^2}{\frac{1}{2}\rho S}$

$A = \frac{1}{2}\rho S C_{x_0}$

Thrust and Drag vs. velocity.
For a given mass and altitude:

[P] $F - D = mg\gamma$



Changing velocity changes the thrust balance by (mostly) changing drag. Therefore

δ_m, by controlling velocity (and drag) also controls γ

Let's look at how pitch control acts on the slope of the aircraft by taking the example of pulling on the stick (positive moment, nose-up, $\delta_m < 0$) from a steady-state, horizontal flight condition::

Apply $\delta_m < 0$ (pull on stick)

⇓

[M] increase angle of attack α

⇓

$mg = \frac{1}{2}\rho S V^2 C_z$ (↓ decrease, ↑ increases) [S] decrease velocity V

low velocity R_2 / R_1 (high velocity)

$F - D < 0$

$mg\gamma < 0$

$\gamma < 0$

Aircraft descending

2nd regime

$F - D > 0$

$mg\gamma > 0$

$\gamma > 0$

Aircraft ascending

1st regime

8 Models for forces and moment coefficients

In practice, we will use simple linear models to calculate the lift, drag, and pitching moment coefficients of an aircraft based on the aircraft geometry and the relevant controls. These models are extensions from the ones used in MAE 3050 and mentioned in a previous chapter, so that they can account for changes in controls (throttle and pitch) and configurations (deployment of moving surfaces such as flaps or spoilers).

- **Drag coefficient:**

$$C_x = C_{x_0} + k C_z^2 + \Delta C_{x_0}$$

(ΔC_{x_0}: captures effects of landing gear, slats/flaps, spoilers ...)

- **Lift coefficient:**

$$C_z = C_{z_\alpha} \bar{\alpha} + C_{z_{\delta_m}} \delta_m + \Delta C_{z_{sp}}$$

($\bar{\alpha} = \alpha - \alpha_{zero\,lift}$; $\Delta C_{z_{sp}}$: spoilers)

- **Pitching moment coefficient:**

$$C_m = 0 = C_{m_0} + C_{m_\alpha} \bar{\alpha} + C_{m_{\delta_m}} \delta_m + \Delta C_{m_0}$$

($C_m = 0$: at steady state; C_{m_0}: moment coefficient at zero lift; $\bar{\alpha} = (\alpha - \alpha_{zl})$; ΔC_{m_0}: when deploying slats/flaps etc ...)

When those models are used in a case study, the relevant contributions and their values will be clearly stated.